Replay-Aware CVRP Benchmark Report

Overview

- Condition count: 4.
- Best final transfer gap: cvrp_failure_replay.
- Best CVRPLIB holdout gap: cvrp_no_replay.
- Best synthetic holdout gap: cvrp_random_replay.

Run Metadata

- run_name: run_20260513_194251_n
- started_at_local: 2026-05-13 19:42:51
- finished_at_local: 2026-05-13 19:45:45
- duration_hhmm: 00:03
- duration_seconds: 173.659
- seed_offset: 13000
- replicate_label: n
- judge_status: enabled
- judge_provider: openai
- judge_model: gpt-4.1-mini

Condition Comparison

Condition	Replay	Selection	Compression	Final CVRPLIB Gap	Final Synthetic Gap	Final Transfer Gap	Mean Accepted Novelty	Mean Complexity	Adaptation Efficiency
cvrp_no_replay	none	score_only	False	0.212727	0.05147	0.141057	0.0	0.78	0.0
cvrp_random_replay	random	score_only	False	0.246889	0.005376	0.13955	0.890836	0.58	0.002225
cvrp_failure_replay	failure	score_only	False	0.219278	0.005376	0.12421	0.814103	0.78	0.035504
cvrp_failure_replay_compression	failure	novelty_gate	True	0.22301	0.061086	0.151044	0.903226	0.78	-0.015134

Condition Notes

cvrp_no_replay

- Replay mode: none with selection mode score_only.
- Final transfer gaps: CVRPLIB 0.212727, synthetic 0.05147, combined 0.141057.

- Accepted-epoch count 1, mean accepted code novelty 0.0, and final complexity 0.78.
- Adaptation efficiency 0.0 and archive sizes {'worst_cases': 5, 'failure_cases': 5, 'adversarial_layouts': 4} / elite 1.
- Panel heldout_cvrplib mean gap 0.212727 across 5 instances; family means: A=0.165794, B=0.194209, E=0.21881, P=0.290614.
- Panel synthetic_holdout mean gap 0.05147 across 4 instances; family means: alternating_belt=0.049043, clustered_demand=0.0, corridor_split=0.0, radial_heavy=0.156836.
- Worst recent training cases: [{"best_known_cost": 784, "cost": 1088, "family": "A", "name": "A-n32-k5", "optimality_gap": 0.387755}, {"best_known_cost": 937, "cost": 1245, "family": "A", "name": "A-n44-k6", "optimality_gap": 0.328709}, {"best_known_cost": 458, "cost": 585, "family": "P", "name": "P-n40-k5", "optimality_gap": 0.277293}]

cvrp_random_replay

- Replay mode: random with selection mode score_only.
- Final transfer gaps: CVRPLIB 0.246889, synthetic 0.005376, combined 0.13955.
- Accepted-epoch count 2, mean accepted code novelty 0.890836, and final complexity 0.58.
- Adaptation efficiency 0.002225 and archive sizes {'worst_cases': 5, 'failure_cases': 5, 'adversarial_layouts': 4} / elite 1.
- Panel heldout_cvrplib mean gap 0.246889 across 5 instances; family means: A=0.143979, B=0.248809, E=0.285988, P=0.306859.
- Panel synthetic_holdout mean gap 0.005376 across 4 instances; family means: alternating_belt=0.021505, clustered_demand=0.0, corridor_split=0.0, radial_heavy=0.0.
- Worst recent training cases: [{"best_known_cost": 458, "cost": 591, "family": "P", "name": "P-n40-k5", "optimality_gap": 0.290393}, {"best_known_cost": 937, "cost": 1063, "family": "A", "name": "A-n44-k6", "optimality_gap": 0.134472}, {"best_known_cost": 784, "cost": 860, "family": "A", "name": "A-n32-k5", "optimality_gap": 0.096939}]

cvrp_failure_replay

- Replay mode: failure with selection mode score_only.
- Final transfer gaps: CVRPLIB 0.219278, synthetic 0.005376, combined 0.12421.
- Accepted-epoch count 2, mean accepted code novelty 0.814103, and final complexity 0.78.
- Adaptation efficiency 0.035504 and archive sizes {'worst_cases': 5, 'failure_cases': 5, 'adversarial_layouts': 4} / elite 2.
- Panel heldout_cvrplib mean gap 0.219278 across 5 instances; family means: A=0.086387, B=0.211766, E=0.209213, P=0.377256.
- Panel synthetic_holdout mean gap 0.005376 across 4 instances; family means: alternating_belt=0.021505, clustered_demand=0.0, corridor_split=0.0, radial_heavy=0.0.
- Worst recent training cases: [{"best_known_cost": 458, "cost": 608, "family": "P", "name": "P-n40-k5", "optimality_gap": 0.327511}, {"best_known_cost": 937, "cost": 1219, "family": "A", "name": "A-n44-k6", "optimality_gap": 0.300961}, {"best_known_cost": 672, "cost": 807, "family": "B", "name": "B-n31-k5", "optimality_gap": 0.200893}]

cvrp_failure_replay_compression

- Replay mode: failure with selection mode novelty_gate.
- Final transfer gaps: CVRPLIB 0.22301, synthetic 0.061086, combined 0.151044.

- Accepted-epoch count 2, mean accepted code novelty 0.903226, and final complexity 0.78.
- Adaptation efficiency -0.015134 and archive sizes {'worst_cases': 5, 'failure_cases': 5, 'adversarial_layouts': 4} / elite 2.
- Panel heldout_cvrplib mean gap 0.22301 across 5 instances; family means: A=0.206806, B=0.322363, E=0.120921, P=0.142599.
- Panel synthetic_holdout mean gap 0.061086 across 4 instances; family means: alternating_belt=0.10097, clustered_demand=0.043468, corridor_split=0.099906, radial_heavy=0.0.
- Worst recent training cases: [{"best_known_cost": 937, "cost": 1219, "family": "A", "name": "A-n44-k6", "optimality_gap": 0.300961}, {"best_known_cost": 458, "cost": 588, "family": "P", "name": "P-n40-k5", "optimality_gap": 0.283843}, {"best_known_cost": 672, "cost": 807, "family": "B", "name": "B-n31-k5", "optimality_gap": 0.200893}]

Judge Appendix

Summary of CVRP Benchmark Results by Replay Condition

Condition	Final CVRPLIB Gap	Final Synthetic Gap	Final Transfer Gap	Code Novelty (Mean)	Compression Pressure	Replay Mode	Notes on Transfer & Optimality
cvrp_no_replay	**0.213**	0.051	0.141	0.0	No	None	Best CVRPLIB gap (heldout), moderate transfer gap; no code novelty.
cvrp_random_replay	0.247	**0.0054**	0.140	0.89	No	Random	Best synthetic gap, transfer gap close to no replay; high code novelty but worse heldout optimality.
cvrp_failure_replay	0.219	0.0054	**0.124**	0.81	No	Failure	Best transfer gap; synthetic gap equal to random replay; intermediate heldout gap; substantial code novelty.
cvrp_failure_replay_compression	0.223	0.061	0.151	0.90	Yes	Failure + Compression-aware	Highest synthetic and transfer gaps, close heldout gap to failure replay; high code novelty but negative adaptation efficiency.

Interpretation (Conservative, Metric-Focused)

- **Heldout CVRPLIB gap (main optimality indicator):**
- **No replay condition ("cvrp_no_replay") achieved the best optimality on heldout CVRPLIB instances (0.213 gap).**
- Replay-based methods show **worse heldout gaps**, with random replay the worst (0.247), failure replay slightly better (0.219), and compression replay close (0.223).
- This suggests replay mechanisms do **not improve heldout CVRPLIB performance**; in fact, they degrade it mildly.

- **Synthetic holdout gap (generalization to synthetic scenarios):**
- **Replay conditions with failure and random replay yielded very low synthetic gaps (~0.0054), far superior to no replay (0.051) and compression replay (0.061).**

- This indicates replay improves synthetic transfer optimality, especially failure and random replay modes.

- **Mean transfer gap (average generalization):**
- Best transfer gap is from **failure replay (0.124), slightly better than random replay (0.140) and no replay (0.141).**
- Compression replay shows the worst transfer gap (0.151).
- Failure replay thus offers **the best transfer generalization** overall.

- **Code novelty:**
- No replay condition has **zero code novelty**.
- Replay conditions show high novelty (0.81–0.90), with compression replay highest (0.90).
- Despite this code novelty increase, the **heldout CVRPLIB optimality did not improve (or even worsened).**
- This implies increased code novelty does **not correspond to better optimality on heldout CVRPLIB**.

- **Compression pressure:**
- Only the failure replay compression-aware condition has compression pressure; resulted in **lower adaptation efficiency (-0.015) and worse transfer gap,** despite high code novelty.
- This suggests compression pressure may harm transfer and adaptation in this setting.

- **Adaptation Efficiency:**
- Highest in failure replay (0.0355), positive but low in random replay (0.0022), zero in no replay, and negative in failure replay compression.
- Failure replay may enable better adaptation but does not lead to better heldout optimality.

Conclusions

- The **no replay baseline achieves the best CVRPLIB heldout optimality** and moderate transfer, despite zero code novelty and no replay.
- Both **random and failure replay improve synthetic holdout performance significantly**, with synthetic gap dropping from ~0.05 (no replay) to ~0.005.
- **Failure replay leads to the best overall transfer gap**, outperforming no replay and random replay in transfer generalization, while maintaining moderate CVRPLIB gaps.
- **Code novelty increases substantially under replay conditions but does not translate into better heldout CVRPLIB optimality,** indicating novelty is not equivalent to algorithmic improvement in this context.
- Compression-aware failure replay introduces negative adaptation efficiency and worsens transfer despite very high novelty, suggesting compression pressure is detrimental here.
- Replay mode matters: **failure replay offers the best transfer improvement, albeit with some degradation in heldout optimality** compared to no replay.
- Narrative speculation about code novelty or replay being beneficial should be tempered—**metrics show replay improves synthetic transfer but reduces heldout optimality.**

Recommendation

- Prioritize no replay for best heldout CVRPLIB gap if final optimality is primary.
- Use failure replay when transfer generalization is critical and slight optimality degradation is acceptable.
- Avoid compression-aware replay due to negative adaptation effects.
- Treat code novelty increases cautiously; they do not guarantee algorithmic or performance improvements.